

STAT525 HOMEWORK #7

1. KNNL Problem 11.7 (part (b) is optional) (*Non-STAT Majors*)
2. KNNL Problem 11.16
3. KNNL Problem 16.2
4. KNNL Problem 16.5
5. KNNL Problem 16.47 (*Non-STAT Majors*)

For the remainder of the assignment, use the data from problem 16.11 described on page 725 of the text.

6. Give a table of sample sizes, means, and standard deviations for the six different filling machines.
7. Statistically examine the question of whether or not the six machines place the same amount of fill into the cartons. Write a model for this analysis, state the null and alternative hypothesis in terms of your model parameters (cell-based or factor effects), give the test statistic with degrees of freedom, the P-value, and your conclusion.
8. Examine the residuals of this analysis to make sure the assumptions are not violated. Display (and comment on) the plots and/or tests you use to do this.

1. KNNL Problem 11.7 (part (b) is optional)

- *11.7. **Machine speed.** The number of defective items produced by a machine (Y) is known to be linearly related to the speed setting of the machine (X). The data below were collected from recent quality control records.

i :	1	2	3	4	5	6	7	8	9	10	11	12
X_i :	200	400	300	400	200	300	300	400	200	400	200	300
Y_i :	28	75	37	53	22	58	40	96	46	52	30	69

- a. Fit a linear regression function by ordinary least squares, obtain the residuals, and plot the residuals against X . What does the residual plot suggest?

```

25 /* --- Fit OLS regression --- */
26 proc reg data=machinespeed;
27     model y = x;
28     output out=residdata r=residual p=predicted;
29 run;
30
31 proc print data=residdata;
32     var x y predicted residual;
33     title "Observed, Predicted, and Residual Values";
34 run;
35
36 /* --- Plot residuals against X --- */
37 symbol v=circle i=none;
38 proc gplot data=residdata;
39     plot residual*x / vaxis=axis1 haxis=axis2 frame;
40     title "Residual Plot for Machine Speed vs Number of Defectives";
41     title2 "Checking for Unequal Error Variance (Ch. 11 Concept)";
42 run;
43 quit;

```

The REG Procedure
Model: MODEL1
Dependent Variable: y

OLS

Number of Observations Read	12
Number of Observations Used	12

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	2812.50000	2812.50000	12.14	0.0059
Error	10	2316.50000	231.65000		
Corrected Total	11	5129.00000			

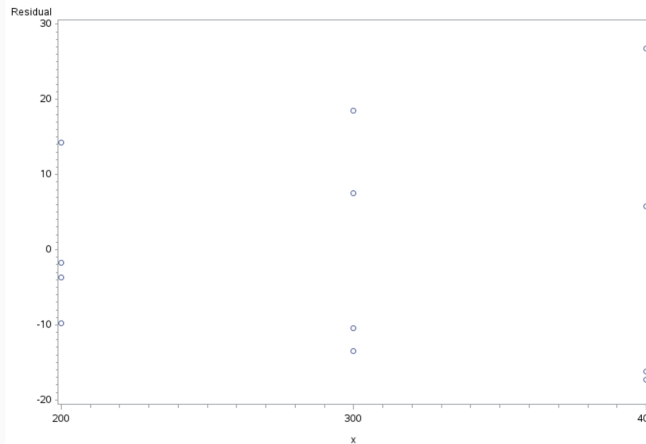
Root MSE	15.22005	R-Square	0.5484
Dependent Mean	50.50000	Adj R-Sq	0.5032
Coeff Var	30.13872		

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	-5.75000	16.73052	-0.34	0.7382
x	1	0.18750	0.05381	3.48	0.0059

Observed, Predicted, and Residual Values

Obs	x	y	predicted	residual
1	200	28	31.75	-3.75
2	400	75	69.25	5.75
3	300	37	50.50	-13.50
4	400	53	69.25	-16.25
5	200	22	31.75	-9.75
6	300	58	50.50	7.50
7	300	40	50.50	-10.50
8	400	96	69.25	26.75
9	200	46	31.75	14.25
10	400	52	69.25	-17.25
11	200	30	31.75	-1.75
12	300	69	50.50	18.50

Residual Plot for Machine Speed vs Number of Defectives
Checking for Unequal Error Variance (Ch. 11 Concept)



↑ spread of e_i

$$\hat{Y} = -5.75 + 0.1875X$$

- Residual Plot suggests Heteroscedasticity; error var. ↑ with X (Machine Speed)

- when $X=200 \rightarrow e_i$ around -3 to +14
- when $X=300 \rightarrow e_i$ up to +18
- when $X=400 \rightarrow e_i$ -17 to +27

- b. Conduct the Breusch-Pagan test for constancy of the error variance, assuming $\log_e \sigma_i^2 = \gamma_0 + \gamma_1 X_i$; use $\alpha = .10$. State the alternatives, decision rule, and conclusion.

(Skipped; optional)

- c. Plot the squared residuals against X . What does the plot suggest about the relation between the variance of the error term and X ?

```

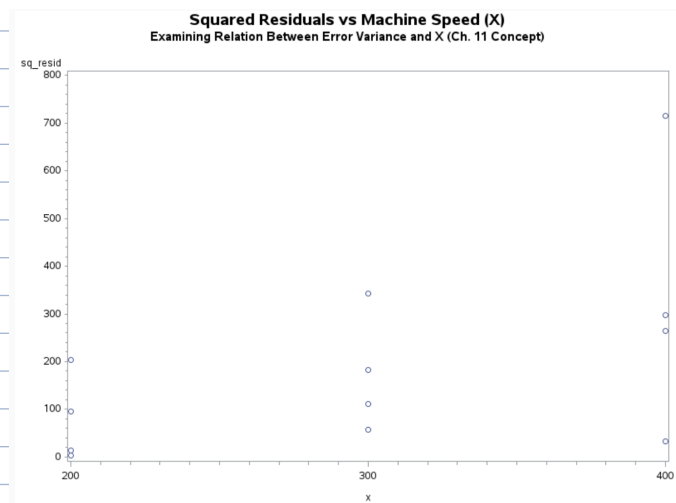
44 /* Use residual dataset from part (a) */
45 data resid_sqr;
46     set residdata;
47     sq_resid = residual**2;
48 run;
49
50 /* --- Plot squared residuals against X --- */
51 symbol v=circle i=none;
52 proc gplot data=resid_sqr;
53     plot sq_resid*x / vaxis=axis1 haxis=axis2 frame;
54     title "Squared Residuals vs Machine Speed (X)";
55     title2 "Examining Relation Between Error Variance and X (Ch. 11 Concept)";
56 run;

```

↑ Spread of e_i^2

same pattern as (a):

- Residual² Plot suggests Heteroscedasticity; error var. ↑ with X (Machine Speed)
- Error var. is proportional to Machine Speed.



- d. Estimate the variance function by regressing the squared residuals against X , and then calculate the estimated weight for each case using (11.16b).

$$w_i = \frac{1}{\hat{v}_i}$$

where $\hat{v}_i$ is fitted value from variance function

(11.16b)

```

58 /* 1) Squared residuals from part (a) */
59 data resid_sqr; set residdata; sq_resid = residual**2; run;
60
61 /* 2) Regress e^2 on X to get \hat{\sigma}^2_i */
62 proc reg data=resid_sqr;
63     model sq_resid = x;
64     output out=varfit p=vhat; /* vhat = \hat{\sigma}^2_i */
65 run; quit;
66
67 /* 3) Compute weights w_i = 1 / \hat{\sigma}^2_i (guard against tiny/negatives) */
68 data wts;
69     set varfit;
70     if vhat < 1e-6 then vhat = 1e-6;
71     w = 1 / vhat;
72 run;
73
74 proc print data=wts noobs;
75     var x y residual sq_resid vhat w;
76     title "Weights via (11.16b): w_i = 1 / \hat{\sigma}^2_i";
77 run;

```

Weights via (11.16b): $w_i = 1 / \hat{\sigma}^2_i$

x	y	residual	sq_resid	vhat	w
200	28	-3.75	14.063	68.667	0.014563
400	75	5.75	33.063	317.417	0.003150
300	37	-13.50	182.250	193.042	0.005180
400	53	-16.25	264.063	317.417	0.003150
200	22	-9.75	95.063	68.667	0.014563
300	58	7.50	56.250	193.042	0.005180
300	40	-10.50	110.250	193.042	0.005180
400	96	26.75	715.563	317.417	0.003150
200	46	14.25	203.063	68.667	0.014563
400	52	-17.25	297.563	317.417	0.003150
200	30	-1.75	3.063	68.667	0.014563
300	69	18.50	342.250	193.042	0.005180

• Lower speed → Larger weight (Var. ↓)

The REG Procedure
Model: MODEL1
Dependent Variable: sq_resid

Number of Observations Read	12
Number of Observations Used	12

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	123753	123753	3.91	0.0762
Error	10	316609	31661		
Corrected Total	11	440362			

Root MSE	177.93503	R-Square	0.2810
Dependent Mean	193.04167	Adj R-Sq	0.2091
Coeff Var	92.17442		

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	-180.08333	195.59369	-0.92	0.3789
x	1	1.24375	0.62910	1.98	0.0762

$$\hat{e}_i^2 = \hat{v}_i = -180.083 + 1.244 X_i$$

- e. Using the estimated weights, obtain the weighted least squares estimates of β_0 and β_1 . Are the weighted least squares estimates similar to the ones obtained with ordinary least squares in part (a)?

```

79 /* OLS (refit just to capture a tidy table) */
80 ods output ParameterEstimates=ols_pe;
81 proc reg data=machinespeed;
82     model y = x;
83 run; quit;
84
85 /* WLS using weights from part (d) */
86 ods output ParameterEstimates=wls_pe;
87 proc reg data=wts;
88     model y = x / clb;
89     weight w; /* w_i = 1 / vhat_i */
90     output out=wlsout r=res_w p=pred_w;
91     title "WLS Fit using w_i = 1/vhat_i (Eq. 11.16b)";
92 run; quit;
93
94 /* Combine and label for side-by-side comparison */
95 data ols_pe; set ols_pe; Method = "OLS"; keep Method Variable Estimate StdErr tValue Probt; run;
96 data wls_pe; set wls_pe; Method = "WLS"; keep Method Variable Estimate StdErr tValue Probt; run;
97
98 data compare;
99     set ols_pe wls_pe;
100 run;
101
102 proc print data=compare noobs;
103     title "Comparison of OLS and WLS Estimates";
104 run;

```

WLS Fit using $w_i = 1/\hat{v}_i$ (Eq. 11.16b) **WLS**

The REG Procedure
Model: MODEL1
Dependent Variable: y

Number of Observations Read	12
Number of Observations Used	12

Weight: w

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	17.20088	17.20088	13.99	0.0038
Error	10	12.29657	1.22966		
Corrected Total	11	29.49745			

Root MSE	1.10890	R-Square	0.5831
Dependent Mean	41.07272	Adj R-Sq	0.5414
Coeff Var	2.69984		

Parameter Estimates						
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t	95% Confidence Limits
Intercept	1	-6.23322	13.16843	-0.47	0.6461	-35.57431 23.10787
x	1	0.18911	0.05056	3.74	0.0038	0.07645 0.30177

• WLS estimates $\hat{\beta}_0 = -6.23$, $\hat{\beta}_1 = 0.1891$, very similar to OLS.

• However, the WLS SE are ↓, so its t-value is slightly bigger, so WLS slope is more significant.

• Meaning WLS successfully corrected for non-constant variance, yielding more efficient estimates.

Comparison of OLS and WLS Estimates

Variable	Estimate	StdErr	tValue	Probt	Method
Intercept	-5.75000	16.73052	-0.34	0.7382	OLS
x	0.18750	0.05381	3.48	0.0059	OLS
Intercept	-6.23322	13.16843	-0.47	0.6461	WLS
x	0.18911	0.05056	3.74	0.0038	WLS

- f. Compare the estimated standard deviations of the weighted least squares estimates b_{w0} and b_{w1} in part (e) with those for the ordinary least squares estimates in part (a). What do you find?

```

106 /* --- Create summary datasets with OLS and WLS estimates --- */
107 data ols_results;
108     input Method $ Intercept SE_Intercept Slope SE_Slope;
109     datalines;
110 OLS -5.7500 16.7305 0.1875 0.05381
111 ;
112 run;
113
114 data wls_results;
115     input Method $ Intercept SE_Intercept Slope SE_Slope;
116     datalines;
117 WLS -6.2332 13.1684 0.1891 0.05056
118 ;
119 run;
120
121 /* --- Combine both into one dataset --- */
122 data compare_SE;
123     set ols_results wls_results;
124 run;
125
126 /* --- Calculate percent reduction in SEs --- */
127 proc sql;
128     create table se_change as
129     select
130         a.SE_Intercept as SE_Intercept_OLS,
131         b.SE_Intercept as SE_Intercept_WLS,
132         100*(a.SE_Intercept - b.SE_Intercept)/a.SE_Intercept as Pct_Reduction_Intercept format=6.2,
133         a.SE_Slope as SE_Slope_OLS,
134         b.SE_Slope as SE_Slope_WLS,
135         100*(a.SE_Slope - b.SE_Slope)/a.SE_Slope as Pct_Reduction_Slope format=6.2
136     from ols_results a, wls_results b;
137 quit;
138
139 /* --- Display comparison --- */
140 title "Comparison of Standard Errors (OLS vs WLS)";
141 proc print data=se_change noobs label;
142     label SE_Intercept_OLS = "OLS SE(b0)"
143           SE_Intercept_WLS = "WLS SE(b0)"
144           Pct_Reduction_Intercept = "% Reduction in SE(b0)"
145           SE_Slope_OLS = "OLS SE(b1)"
146           SE_Slope_WLS = "WLS SE(b1)"
147           Pct_Reduction_Slope = "% Reduction in SE(b1)";
148 run;

```

Comparison of Standard Errors (OLS vs WLS)

OLS SE(b0)	WLS SE(b0)	% Reduction in SE(b0)	OLS SE(b1)	WLS SE(b1)	% Reduction in SE(b1)
16.7305	13.1684	21.29	0.05381	0.05056	6.04

• SE in WLS are much smaller than SE in OLS.

• Meaning WLS is more efficient and the variance-stabilizing weights successfully corrected for non-constant variance.

g. Iterate the steps in parts (d) and (e) one more time. Is there a substantial change in the estimated regression coefficients? If so, what should you do?

```

1 /* Step 1: Use WLS residuals from part (e) */
2 data iter1;
3   set wlsout;                      /* residuals from previous WLS */
4   sq_resid_wls = res_w**2;
5 run;
6
7 /* Step 2: Refit variance model e_wls^2 ~ X */
8 proc reg data=iter1;
9   model sq_resid_wls = x;
10  output out=iter_varfit p=vhat2;    /* new variance fit */
11 run; quit;
12
13 /* Step 3: Compute new weights w2_i = 1 / vhat2_i */
14 data iter_wts;
15   set iter_varfit;
16   if vhat2 < 1e-6 then vhat2 = 1e-6;
17   w2 = 1 / vhat2;
18 run;
19
20 /* Step 4: Refit WLS again with updated weights */
21 proc reg data=iter_wts;
22   model y = x;
23   weight w2;
24   title "Iterated WLS (2nd round) using new weights";
25 run; quit;

```

- Only $\hat{\beta}_0$ changes little from -6.23322 v.s. -6.23349
- $\hat{\beta}_1$ stays the same

• Repeating WLS produced no meaningful change to $\hat{\beta}_0, \hat{\beta}_1$.

• The weights based on $w_i = 1/\hat{v}_i$ have already stabilized, so the model has converged and the first WLS can be accepted as final.

WLS Fit using $w_i = 1/\hat{v}_i$ (Eq. 11.16b) WLS

The REG Procedure
Model: MODEL1
Dependent Variable: y

Number of Observations Read	12
Number of Observations Used	12

Weight: w

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	17.20088	17.20088	13.99	0.0038
Error	10	12.29657	1.22966		
Corrected Total	11	29.49745			

Root MSE	1.10890	R-Square	0.5831
Dependent Mean	41.07272	Adj R-Sq	0.5414
Coeff Var	2.89984		

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t	95% Confidence Limits
Intercept	1	-6.23322	13.16843	-0.47	0.6461	-35.57431 23.10787
x	1	0.18911	0.05056	3.74	0.0038	0.07645 0.30177

Iterated WLS (2nd round) using new weights WLS-2

The REG Procedure
Model: MODEL1
Dependent Variable: y

Number of Observations Read	12
Number of Observations Used	12

Weight: w2

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	17.20258	17.20258	13.99	0.0038
Error	10	12.29921	1.22992		
Corrected Total	11	29.50179			

Root MSE	1.10902	R-Square	0.5831
Dependent Mean	41.06573	Adj R-Sq	0.5414
Coeff Var	2.70059		

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	-6.23349	13.16725	-0.47	0.6461
x	1	0.18911	0.05057	3.74	0.0038

2. KNNL Problem 11.16

11.16. Refer to **Machine speed** Problem 11.7. Demonstrate numerically that the weighted least squares estimates obtained in part (e) are identical to those obtained when using transformation (11.23) and ordinary least squares. We premultiply the terms on both sides of regression model (11.21) by $W^{1/2}$ and obtain:

$$W^{1/2}Y = W^{1/2}X\beta + W^{1/2}\epsilon \quad (11.23)$$

```

179 /* 1) Keep WLS estimates to compare */
180 ods output ParameterEstimates=wls_pe;
181 proc reg data=wts;
182   model y = x;
183   weight w;
184   title "WLS (reference)";
185 run; quit;
186
187 /* 2) Build transformed variables: y*sqrt(w), x*sqrt(w), and the transformed
188   intercept column iw = sqrt(w). Then fit OLS with NO intercept. */
189 data trans;
190   set wts;
191   sw = sqrt(w);          /* sqrt of weight */
192   yw = sw * y;           /* W^{1/2}Y */
193   xw = sw * x;           /* W^{1/2}X for X */
194   iw = sw;               /* W^{1/2} * 1 (intercept column) */
195 run;
196
197 ods output ParameterEstimates=trans_pe;
198 proc reg data=trans;
199   model yw = iw xw / noint; /* OLS on transformed system, NO intercept */
200   title "OLS on transformed system (Eq. 11.23)";
201 run; quit;
202
203 /* 3) Side-by-side comparison */
204 data wls_pe; set wls_pe; Method="WLS"; keep Method Variable Estimate; run;
205 data trans_pe; set trans_pe; Method="Trans-OLS"; keep Method Variable Estimate; run;
206
207 data compare_1116; set wls_pe trans_pe; run;
208
209 proc print data=compare_1116 noobs;
210   title "Compare coefficients: WLS vs OLS on W^{1/2}-transformed system";
211 run;

```

Compare coefficients: WLS vs OLS on $W^{1/2}$ -transformed system

Variable	Estimate	Method
Intercept	-6.23322	WLS
x	0.18911	WLS
iw	-6.23322	Tra
xw	0.18911	Tra

• The numerical equality demonstrates the theoretical equivalence of WLS and OLS regression on transformed data.

3. KNNL Problem 16.2

- 16.2. A market researcher, having collected data on breakfast cereal expenditures by families with 1, 2, 3, 4, and 5 children living at home, plans to use an ordinary regression model to estimate the mean expenditures at each of these five family size levels. However, the researcher is undecided between fitting a linear or a quadratic regression model, and the data do not give clear evidence in favor of one model or the other. A colleague suggests: "For your purposes you might simply use an ANOVA model." Is this a useful suggestion? Explain.

Yes, using an ANOVA model is a useful suggestion.

The linear and quadratic regression models both impose specific functional forms on the relationship between family size and mean expenditure — the linear model assumes a straight-line increase, and the quadratic model assumes a curved trend. However, the data provide no clear evidence favoring either form.

By contrast, the ANOVA model treats family size as a categorical factor rather than a continuous variable. This allows each family-size level (1, 2, 3, 4, and 5 children) to have its own mean, with no assumption about linearity or curvature. As a result, the ANOVA model provides the most flexible and assumption-free method for estimating the mean expenditure at each family size level.

Although this approach uses more parameters and may be slightly less efficient if the true relationship is linear, it is more robust when the functional form is uncertain. Therefore, the colleague's suggestion is appropriate and practical for the researcher's purpose.

4. KNNL Problem 16.5

- 16.5. In a study of length of hospital stay (in number of days) of persons in four income groups, the parameters are as follows: $\mu_1 = 5.1$, $\mu_2 = 6.3$, $\mu_3 = 7.9$, $\mu_4 = 9.5$, $\sigma = 2.8$. Assume that ANOVA model (16.2) is appropriate.

- a. Draw a representation of this model in the format of Figure 16.2.

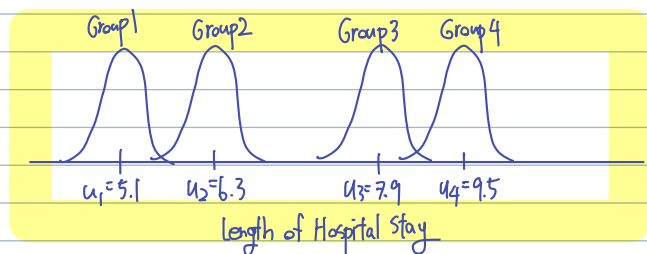
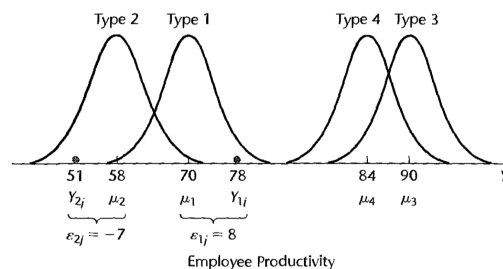


FIGURE 16.2
Analysis of
Variance
Model
Representation
— Incentive
Pay Example.



- b. Suppose 100 persons from each income group are randomly selected for the study. Find $E\{MSTR\}$ and $E\{MSE\}$. Is $E\{MSTR\}$ substantially larger than $E\{MSE\}$ here? What is the implication of this?

$$E\{MSE\} = \sigma^2$$

$$E\{MSE\} = 2.8^2 = 7.84$$

$$u_{\cdot} = \sum n_i u_i / u_{\cdot} = 7.2$$

$$100(5.1)/400$$

$$100(6.3)/400$$

$$100(7.9)/400$$

$$100(9.5)/400$$

$$\sum (u_i - u_{\cdot})^2 = 11.00$$

$$(5.1 - 7.2)^2$$

$$(6.3 - 7.2)^2$$

$$(7.9 - 7.2)^2$$

$$(9.5 - 7.2)^2$$

$$E\{MSTR\} \text{ is much larger than } E\{MSE\}$$

Between Group Variability (diff. among u_i) $\gg$ Within Group Variability (σ^2)

$$E\{MSTR\} = \sigma^2 + \left(\sum n_i (u_i - u_{\cdot})^2 / r - 1 \right)$$

$$u_{\cdot} = \sum n_i u_i / u_{\cdot}$$

$$E\{MSTR\} = 7.84 + 100(11.00)/4 - 1 = 374.5$$

$E\{F\} = E\{MSTR\}/E\{MSE\} \uparrow \uparrow$, Reject H_0 of $u_1 = u_2 = u_3 = u_4$. Income groups have different lengths of stay

- c. If $\mu_2 = 5.6$ and $\mu_3 = 9.0$, everything else remaining the same, what would $E\{MSTR\}$ be? Why is $E\{MSTR\}$ substantially larger here than in part (b) even though the range of the factor level means is the same?

$$u_i = \sum n_{ij} u_{ij} / u_{1j} = 7.3$$

$$\sum (u_{ij} - u_i)^2 = 15.46$$

$$100(5.1)/400$$

$$100(5.6)/400$$

$$100(9.0)/400$$

$$100(9.5)/400$$

$$(5.1-7.3)^2$$

$$(5.6-7.3)^2$$

$$(9.0-7.3)^2$$

$$(9.5-7.3)^2$$

$$E(MSR) c. \ggg E(MSR) b.$$

• $E(MSR)$ depends on sums of (Deviations from Mean)²

• (c) have more extreme means further from group mean.
"further away from the center"

$$E(MSR) = 7.84 + (100(15.46)/4-1) = 523.17$$

5. KNNL Problem 16.47 (Skipped for Non-STAT Majors)

6. Give a table of sample sizes, means, and standard deviations for the six different filling machines.

```
/* Summarize sample size, mean, and SD for each machine */
proc means data=filling n mean stddev;
  class machine;
  var fill;
  title "Table of Sample Sizes, Means, and Standard Deviations for Six Filling Machines";
run;
```

Table of Sample Sizes, Means, and Standard Deviations for Six Filling Machines

The MEANS Procedure

Analysis Variable : fill				
machine	N Obs	N	Mean	Std Dev
1	20	20	0.0735000	0.1925255
2	20	20	0.1905000	0.1854007
3	20	20	0.4600000	0.1646368
4	20	20	0.3655000	0.1653537
5	20	20	0.1250000	0.1727334
6	20	20	0.1515000	0.1734867

7. Statistically examine the question of whether or not the six machines place the same amount of fill into the cartons. Write a model for this analysis, state the null and alternative hypothesis in terms of your model parameters (cell-based or factor effects), give the test statistic with degrees of freedom, the P-value, and your conclusion.

Model: One-way ANOVA

$$Y_{ij} = \mu + \tau_i + \epsilon_{ij}$$

$$H_0: \tau_1 = \tau_2 = \tau_3 = \tau_4 = \tau_5 = \tau_6 = 0$$

$$\mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5 = \mu_6$$

- Y_{ij} : Amount to fill from the j -th obs. from i -th machine.
- μ : overall mean fill level
- τ_i : Effect on the i -th machine ($i=1, \dots, 6$)
- $\epsilon_{ij} \sim N(0, \sigma^2)$, iid.

$$H_a: \text{At least one } \tau_i \neq 0$$

SAS Below:



```

18 proc glm data=filling;
19 class machine;
20 model fill = machine;
21 means machine / hovtest=levene; /* Optional check of equal variances */
22 title "One-Way ANOVA for Fill Amounts by Machine";
23 run;
24 quit;

```

One-Way ANOVA for Fill Amounts by Machine

The GLM Procedure

Class Level Information			
Class	Levels	Values	
machine	6	1 2 3 4 5 6	

Number of Observations Read	120
Number of Observations Used	120

One-Way ANOVA for Fill Amounts by Machine

The GLM Procedure

Dependent Variable: fill

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	5	2.28934667	0.45786933	14.78	<.0001
Error	114	3.53060000	0.03097018		
Corrected Total	119	5.81994667			

R-Square	Coeff Var	Root MSE	fill Mean
0.393362	77.29873	0.175983	0.227667

Source	DF	Type I SS	Mean Square	F Value	Pr > F
machine	5	2.28934667	0.45786933	14.78	<.0001

Source	DF	Type III SS	Mean Square	F Value	Pr > F
machine	5	2.28934667	0.45786933	14.78	<.0001

One-Way ANOVA for Fill Amounts by Machine

The GLM Procedure

Levene's Test for Homogeneity of fill Variance ANOVA of Squared Deviations from Group Means					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
machine	5	0.00142	0.000285	0.27	0.9312
Error	114	0.1224	0.00107		

Test statistic: $F(5, 114) = 14.78, p < 0.0001$

Levene's Test: $F(5, 114) = 0.27, p = 0.93$

- Variances are equal across machines (Levene's Test not significant).

Reject H_0

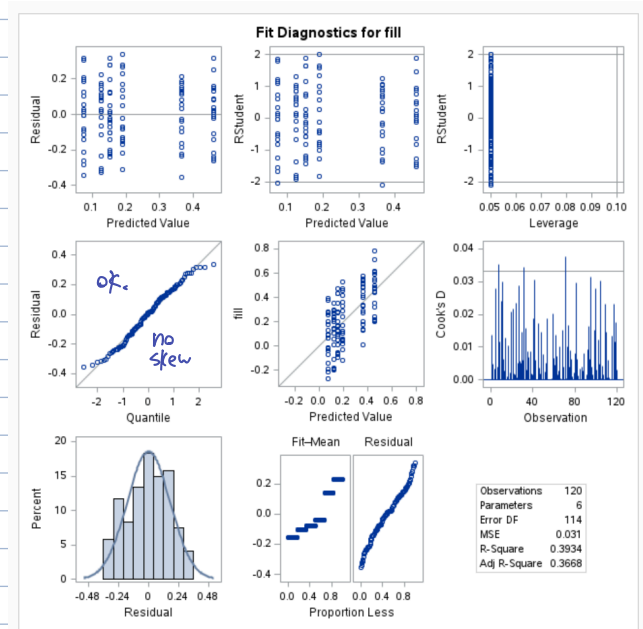
- ANOVA shows highly significant difference among mean fill levels.

- Examine the residuals of this analysis to make sure the assumptions are not violated.
- Display (and comment on) the plots and/or tests you use to do this.

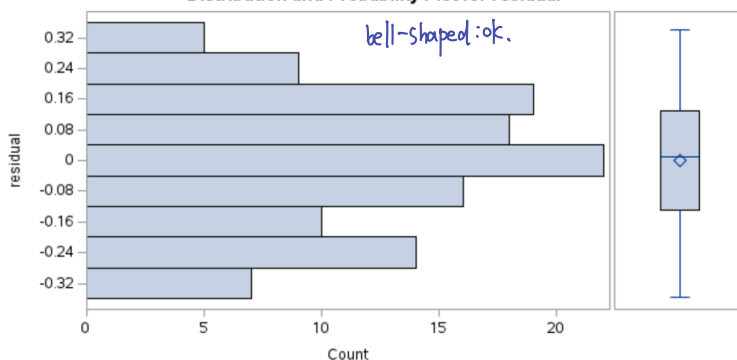
```

3 proc glm data=filling plots=(diagnostics residuals);
4 class machine;
5 model fill = machine;
6 output out=resid_data r=residual p=predicted;
7 title "Residual Diagnostics for Fill Amounts by Machine";
8 run;
9 quit;
10
11 /* Optional: Additional normality tests */
12 proc univariate data=resid_data normal plot;
13 var residual;
14 title "Normality Check for ANOVA Residuals";
15 run;

```



Distribution and Probability Plot for residual



Normality Check for ANOVA Residuals

The UNIVARIATE Procedure Variable: residual

Moments			
N	120	Sum Weights	120
Mean	0	Sum Observations	0
Std Deviation	0.17224665	Variance	0.02966891
Skewness	-0.0920497	Kurtosis	-0.790829
Uncorrected SS	3.5306	Corrected SS	3.5306
Coeff Variation	.	Std Error Mean	0.0157239

Basic Statistical Measures			
Location		Variability	
Mean	0.00000	Std Deviation	0.17225
Median	0.01000	Variance	0.02967
Mode	-0.13050	Range	0.69500
		Interquartile Range	0.25975

Note: The mode displayed is the smallest of 12 modes with a count of 2.

Tests for Location: Mu0=0			
Test	Statistic	p Value	
Student's t	t 0	Pr > t	1.0000
Sign	M 2	Pr >= M	0.7843
Signed Rank	S 41.5	Pr >= S	0.9140

Tests for Normality

Test	Statistic	p Value		
Shapiro-Wilk	W	0.980891	Pr < W	0.0862
Kolmogorov-Smirnov	D	0.056923	Pr > D	>0.1500
Cramer-von Mises	W-Sq	0.068139	Pr > W-Sq	>0.2500
Anderson-Darling	A-Sq	0.482692	Pr > A-Sq	0.2327

→ Fail to Reject : ok.
→ Fail to Reject : ok.
→ Fail to Reject : ok.
→ Fail to Reject : ok.

Quantiles (Definition 5)

Level	Quantile
100% Max	0.33950
99%	0.32000
95%	0.27900
90%	0.21975
75% Q3	0.12925
50% Median	0.01000
25% Q1	-0.13050
10%	-0.23775
5%	-0.29425
1%	-0.34350
0% Min	-0.35550

Extreme Observations

Lowest		Highest	
Value	Obs	Value	Obs
-0.3555	71	0.3065	5
-0.3435	8	0.3165	11
-0.3250	95	0.3185	104
-0.3150	81	0.3200	42
-0.3105	27	0.3395	32

All residual plots and formal tests indicate that the ANOVA model assumptions are satisfied:

Residuals vs fitted show six vertical bands (one per machine), as expected in one-way ANOVA. The spread is comparable across bands, points are centered around 0 with most within ± 2 , and no funnel/curvature is evident—assumptions appear satisfied.

Equal variance holds (Levene $p = 0.93$).

Normality assumption is reasonable (Shapiro-Wilk $p = 0.086 > 0.05$).

Therefore, the ANOVA results from Problem 7 are valid, and the conclusion that at least one filling machine differs in mean fill amount can be trusted.